INCH-POUND

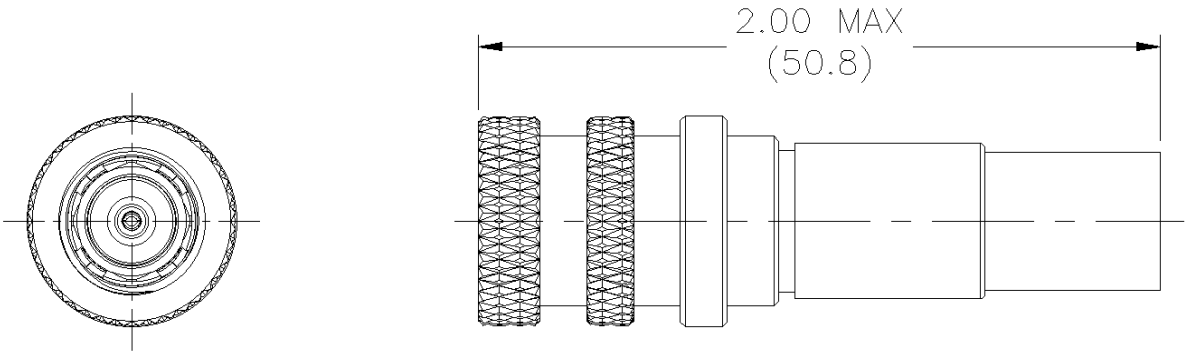
MIL-PRF-49142/16
4 December 2019

PERFORMANCE SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY,
(SERIES TRL, GENDER CHANGEABLE CONTACT , CLASS 2)

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of
this specification sheet and MIL-PRF-49142.

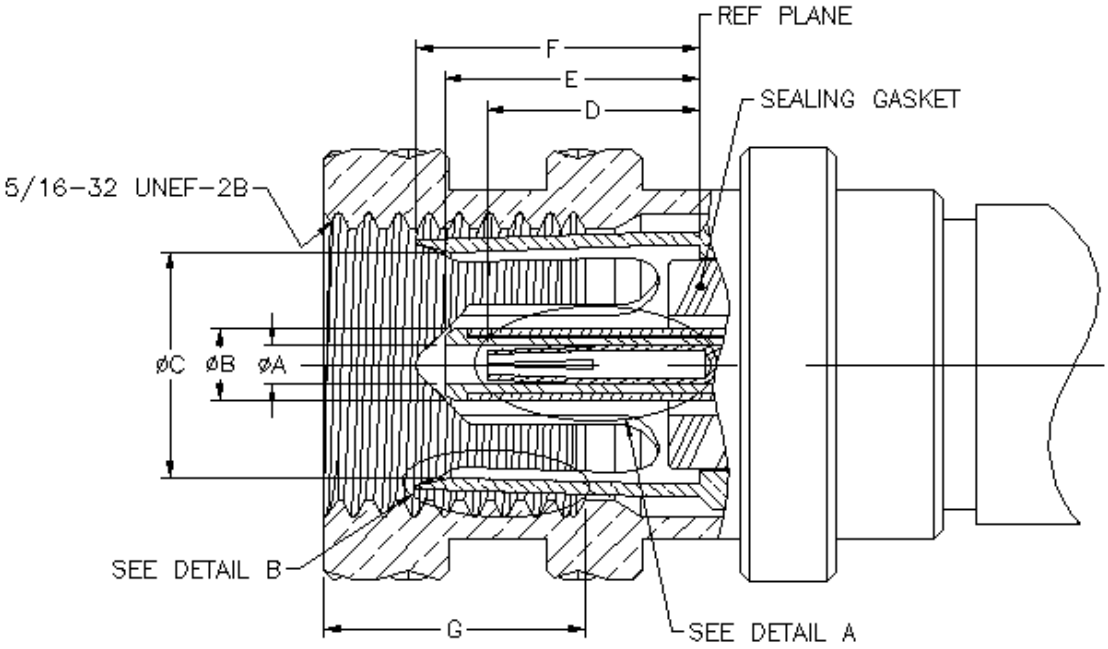


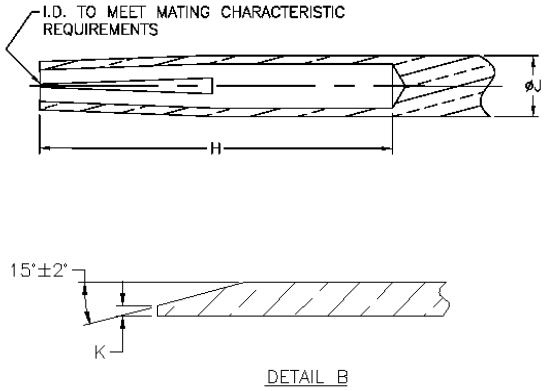

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. Undimensioned pictorial representations for reference only.
3. Wrench flats and hex nuts to accommodate standard wrench openings in accordance with FED-STD-H28.

FIGURE 1. General configuration.



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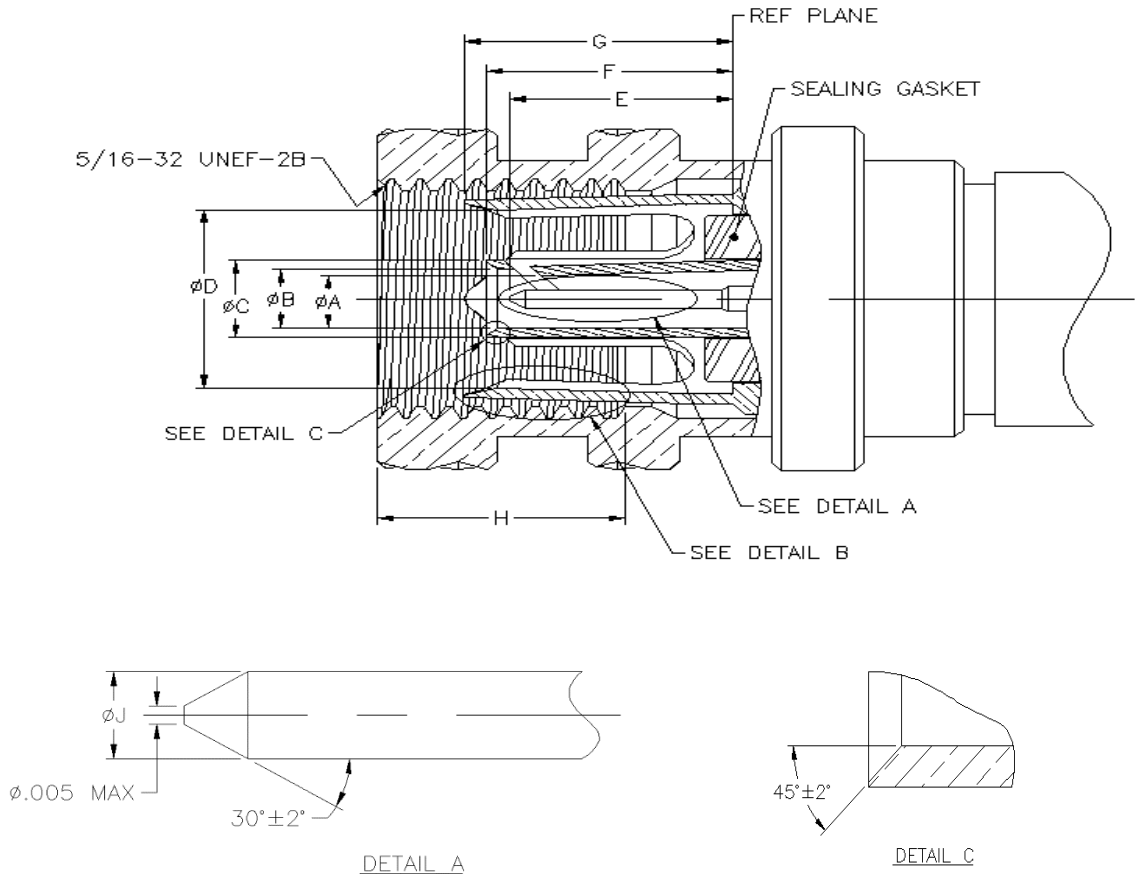

| LTR | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | .038 | .040 | 0.97 | 1.02 |
| B | .072 | .074 | 1.83 | 1.88 |
| C | .225 | .231 | 5.72 | 5.87 |
| D | .188 | .238 | 4.78 | 6.05 |
| E | .234 | .280 | 5.94 | 7.11 |
| F | .281 | .293 | 7.14 | 7.44 |
| G | .220 | - | 5.59 | - |
| H | .222 | - | 5.64 | - |
| J | .038 | .039 | 0.97 | 0.99 |
| K | .001 | .010 | 0.97 | 0.99 |

NOTES:
1. To meet mating characteristics.
2. Dimensions are in inches. Metric equivalents are for reference only.

FIGURE 2. Mating dimensions, pin center contact.



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| LTR | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | See Note 1 | | | |
| B | .075 | .077 | 1.91 | 1.96 |
| C | .100 | .102 | 2.54 | 2.59 |
| D | .225 | .231 | 5.72 | 5.87 |
| E | .209 | .265 | 5.31 | 6.73 |
| F | .245 | .281 | 6.22 | 7.14 |
| G | .281 | .293 | 7.14 | 7.44 |
| H | .220 | - | 5.59 | - |
| J | .023 | .025 | 0.58 | 0.64 |
| K | .001 | .010 | 0.03 | 0.64 |

NOTES:
1. To meet mating characteristics.
2. Dimensions are in inches. Metric equivalents are for reference only.

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FIGURE 3. Mating dimensions, socket center contact.



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ENGINEERING DATA:

Nominal impedance: Non-constant.

Frequency range: 0 to 500 MHz minimum.

Voltage rating:

At sea level: 400 V rms maximum working voltage.

At 70,000 feet (4.437 kPa): 100 V rms maximum working voltage.

Working current: 3 A maximum.

Operating temperature -65°C to +125°C.

Storage temperature: -65°C to +125°C.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Figure 4. Mating connector in accordance with MIL-PRF-49142/15.

Materials:

Insulators: PTFE fluorocarbon in accordance with ASTM D4894 or D4895, or POM in accordance with ASTM D6778.

All other components: In accordance with MIL-PRF-49142.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2.5-inch pounds (.28 Nm) maximum.

Coupling proof torque: 10 inch-pounds (1.13 Nm) minimum.

Mating characteristics: See Figure 2 and Figure 3 for dimensions.

Intermediate contact (socket):

Steel test pin finish: 16 microinches (.406 µm).

Insertion depth: .125 inches (3.17mm) minimum.

Oversize test pin: .075 inches (1.91 mm) diameter minimum.

Number of insertions: 1.

Insertion force test:

Steel test pin diameter: .074 inches (1.88 mm) minimum.



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Insertion force: 1.25 pounds (5.56 N) maximum.

Withdraw force test:

Steel test pin diameter: .072 inches (1.83 mm) maximum.

Withdrawal force: 2.0 ounces (.56 N) minimum.

Center contact (socket):

Steel test pin finish: 16 micorinches (.406 µm).

Insertion depth: .050 inches (3.17mm) minimum.

Oversize test pin: .025 inches (.635 mm) diameter minimum.

Number of insertions: 1.

Insertion force test:

Steel test pin diameter: .0245 inches (.622 mm) minimum.

Insertion force: 1 pounds (4.45 N) maximum.

Withdraw force test:

Steel test pin diameter: .0235 inches (.597 mm) maximum.

Withdrawal force: 0.5 ounces (.14 N) minimum.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: Not applicable.

Insulation resistance:

Center conductor to intermediate conductor: 5,000 megohms minimum.

Intermediate conductor to outer conductor: 5,000 megohms minimum.

Center and intermediate conductor retention: 6 pounds (26.69 N) minimum axial force.

Dielectric withstanding voltage at sea level:

Center conductor to intermediate conductor: 900 V rms minimum.

Intermediate conductor to outer conductor: 900 V rms minimum.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

Shock: Applicable.



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Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

| | Initial | After environment |
|---|---|---|
| Center conductor: | 6.0 | 8.0 |
| Intermediate conductor | 3.0 | 5.0 |
| Outer conductor (nickel plated) | 5.0 | 8.0 |

Corona level:

Altitude: 70,000 feet (4.437 kPa)

Voltage: 250 V rms minimum.

RF high potential withstanding voltage:

Center conductor to intermediate conductor: 900 V rms.

Intermediate conductor to outer conductor: 900 V rms at 5 MHz to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable outside diameter):

.154 inch (3.93 mm) and smaller: Not applicable

.155 - .189 inch (3.94 – 4.82 mm): 10 pounds (44.48 N) minimum.

.190 - .229 inch (4.83 – 5.83 mm): 20 pounds (88.96 N) minimum.

.230 - .249 inch (5.84 – 6.34 mm): 30 pounds (133.45 N) minimum.

.250 inch (6.35 mm) and larger: 40 pounds (177.93 N) minimum.

Coupling mechanism retention force: 100 pounds (444.82 N) minimum.

Rise time degradation: 400 picoseconds maximum (not applicable to connectors using twin conductor cables).

Connector durability: 500 cycles minimum at 12 cycles per minute maximum.

RF leakage: Not applicable.



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Referenced documents. In addition to MIL-PRF-49142, this document references the following:

| | |
|---|---|
| MIL-PRF-49142/15 | ASTM D4894 |
| FED-STD-H28 | ASTM D4895 |
| | ASTM D6778 |

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2019-085)

Review activities:
Army - AR, AT, MI
Navy - AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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